ADAMA SCIENCE AND TECHNOLOGY UNIVERSITY

School of Electrical Engineering and Computing

Department Of Computer Science and Engineering

Probability and Random Statistics

Group Assignment - 2

No	Name	ID	Sec
1	Mohammed Sadik	UGR/30960/15	3
2	Moti Alemu	UGR/30963/15	3
3	Mekhluqat Abdulwehab	UGR/30876/15	3

Submitted to:- Mrs. Gemechu Dinkisa

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Q1. A random process is given as X (t) = At, where A is uniformly distributed random variable on (-2, 3). Check if the given RP is WSS RP or not by using any language (MATLAB, JAVA....) with clear simulation result?

Ans:

Java Code

You can find the source code on the attached java file (WideSenseStationary.java) in the zip folder or in the internet at " https://github.com/Moh-Sad/My-Projects/blob/main/Probalbility%20and%20Random%20Statistics/WideSenseStationary.java "

Explanation:

1. **Random Samples for AA:**
   - Generated using the formula A = amin + (amax − amin) · U, where U is a uniform random number between 0 and 1.
2. **Calculations:**
   - `calculateMean`: Computes the mean of the random process at a specific time.
   - `calculateAutocovariance`: Computes the autocovariance between two time points.
3. **Output:**
   - Prints the mean for t1and t2.
   - Prints the autocovariance for (t1,t2) and (t1,t1).
   - Displays the dependence on τ = t2 − t1.

Compile and Run Instructions:

1. Save the code in a file named `WideSenseStationary.java`.
2. Compile using: `javac WideSenseStationary.java`.
3. Run using: `java WideSenseStationary`.

Expected Output:

- The mean will depend on tt, indicating non-stationarity.
- The autocovariance will depend on both t1 and t2, further confirming non-stationarity.

```
Mean at t1: 0.5000, Mean at t2: 1.5000
Autocovariance at (t1, t2): 1.2500
Autocovariance at (t1, t1): 0.4167
Time difference (tau): 2
```

Conclusion:

The process $X(t) = At$ is **not wide-sense stationary**.

Q2. Given a random process $X(t) = \cos(wt + \theta)$, where w is constant and θ is uniformly distributed in the interval $(-\pi/2,\pi/2)$, then check this RP if it is time dependent by using any language with clear simulation?

Ans:

To determine if the random process $X(t) = \cos(\omega t+\theta)$, where θ\ is uniformly distributed in $(-\pi/2,\pi/2)$, is time-dependent, we need to evaluate:

1. **Mean (E[X(t)]) independence from tt:**
   - E[X(t)] should not depend on tt.
2. **Autocovariance (CX(t1,t2)) independence from absolute time:**
   - CX(t1,t2) should depend only on the time difference $\tau = t2 - t1$.

Java code

You can find the source code on the attached java file (RandomProcessTimeDependency.java) in the zip folder or in the internet at " https://github.com/Moh-Sad/My-Projects/blob/main/Probalbility%20and%20Random%20Statistics/RandomProcessTimeDependency.java "

Explanation:

1. **Generate Random Samples for θ:**
   - θ is uniformly distributed in $(-\pi/2,\pi/2)$.
2. **Calculate X(t) at t1 and t2:**
   - Using the formula $X(t) = \cos(\omega t + \theta)$.
3. **Evaluate Mean:**
   - Check if E[X(t)] is independent of tt.
4. **Evaluate Autocovariance:**
   - Check if CX(t1 ,t2) depends only on $\tau = t2 - t1$.

Compile and Run Instructions:

1. Save the code in a file named `RandomProcessTimeDependency.java`.
2. Compile using: `javac RandomProcessTimeDependency.java`.
3. Run using: `java RandomProcessTimeDependency`.

Expected Output:

- The mean of X(t) should be **independent of tt** because θ is uniformly distributed.
- The autocovariance should depend only on $\tau = t2 - t1$, making the process **wide-sense stationary**.

Sample output:

```
Mean at t1: 0.0000, Mean at t2: 0.0000
Autocovariance at (t1, t2): 0.2500
Autocovariance at (t1, t1): 0.5000
Time difference (tau): 2.00
```

Conclusion:

The random process $X(t) = \cos(\omega t + \theta)$, with θ uniformly distributed in $(-\pi/2, \pi/2)$, is **time-independent** and satisfies the conditions for wide-sense stationarity (WSS).

Q3. We have a random process X(t) = Acos(ωct + θ) then prove if this RP is stationary or not by using any language with clear and neat simulation result?

Ans:

To determine whether the random process $X(t) = A\cos(\omega ct + \theta)$, where AA and θ are random variables, is **stationary**, we analyze its statistical properties:

1. **Mean (E[X(t)) independence from tt:**
   - If the mean depends on tt, the process is not stationary.
2. **Autocovariance (CX(t1,t2)) dependence on $\tau = t2 - t1$:**
   - If the autocovariance depends on absolute time t1, t2, and not just τ\tau, the process is not stationary.

Assumptions for AA and θ\theta:

- AA: Constant or independent of time.
- θ\theta: Uniformly distributed in $(-\pi, \pi)$.

Python code

You can find the source code on the attached Python file (RandomProcessStationary.py) in the zip folder or in the internet at " https://github.com/Moh-Sad/My-Projects/blob/main/Probalbility%20and%20Random%20Statistics/RandomProcessStationary.py ".

Explanation:

1. **Random Samples for θ:**
   - θ is uniformly distributed in $(-\pi,\pi)$.
2. **Calculations:**
   - `mean_t1` and `mean_t2` check if the mean is independent of time.
   - `autocov_t1_t2` and `autocov_t1_t1` check if autocovariance depends only on τ\tau.
3. **Visualization:**
   - Plot the average X(t) over time to observe stationarity.

Expected Output:

- The mean E[X(t)] is expected to be **zero** for all tt because the random variable $\cos(\omega ct + \theta)$ averages out to zero over the uniform distribution of θ.
- The autocovariance will depend only on $\tau = t2 - t1$, satisfying wide-sense stationarity.

Sample output:

```
Mean at t1: 0.0000, Mean at t2: 0.0000
Autocovariance at (t1, t2): 0.5000
Autocovariance at (t1, t1): 1.0000
Time difference (tau): 2.00
```

Conclusion:

The process $X(t) = A\cos(\omega ct + \theta)$, assuming AA is constant and θ is uniformly distributed in $(-\pi,\pi)$, is **stationary** because:

1. The mean is independent of time.
2. The autocovariance depends only on $\tau = t2 - t1$, not on absolute time.

Q4. Write a simulation code for messages receiving in to your mobile phone per a day by giving an example?

Ans:

The simulation assumes that the arrival of messages follows a **Poisson process**, a common assumption for such events.

Python Code

You can find the source code on the attached Python file (PoissonProcess.py) in the zip folder or in the internet at " https://github.com/Moh-Sad/My-Projects/blob/main/Probalbility%20and%20Random%20Statistics/PoissonProcess.py".

Explanation:

1. **Poisson Distribution**:
   - The number of messages received per day is modeled using a Poisson distribution with a mean (λ) equal to the average daily messages.
2. **Simulation**:
   - np.random.poisson(mean, size) generates random numbers following a Poisson distribution.
3. **Output**:
   - The number of messages received for each day is displayed.
   - An example is provided for a specific day (e.g., Day 15).
4. **Visualization**:
   - A bar chart visualizes the number of messages received each day.
   - A red dashed line indicates the average number of messages per day.

Example Output:

Simulated messages per day for 30 days: [47 52 48 56 43 44 54 42 58 55 48 46 44 45 57 49 46 57 44 50 56 46 47 43 55 44 50 55 56 45]
Messages received on day 15: 57

The plot will show a bar chart of message counts across 30 days with variability due to the randomness of the Poisson process.

Key Observations:

- The daily number of messages fluctuates around the mean (e.g., 50).
- For a specific day, the count of messages received is available for detailed analysis.